DG INTERNS HUB — TECHNICAL STUDY WRITEUP

WEEK 4 – SIEM & SECURITY MONITORING: WAZUH PLATFORM

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1. OVERVIEW & PLATFORM CAPABILITIES

Wazuh is an open-source enterprise security monitoring solution combining Extended Detection and Response (XDR) and SIEM capabilities. It provides centralized log management, File Integrity Monitoring (FIM), rootkit detection, vulnerability assessment, regulatory compliance auditing (PCI-DSS, NIST 800-53, CIS), and automated active threat response across Linux, Windows, and macOS endpoints.

2. MANAGER-AGENT ARCHITECTURE

Wazuh utilizes a distributed multi-component architecture:

- **Wazuh Agent:** Lightweight daemon running on target hosts. Collects OS logs, monitors file system changes, scans installed packages, and communicates with the manager over encrypted TCP port 1514.
- **Wazuh Manager:** Central analysis engine. Receives event streams from agents, decodes raw logs, evaluates XML detection rules, triggers active response actions, and passes alerts to the indexer.
- **Wazuh Indexer & Dashboard:** Highly scalable OpenSearch search engine and Kibana-based web GUI used for visualization, threat hunting, and compliance reporting.

3. ENDPOINT LOG COLLECTION CONFIGURATION

Log collection on endpoints is configured within the `ossec.conf` file (located at `/var/ossec/etc/ossec.conf` on Linux or `C:\Program Files (x86)\ossec-agent\ossec.conf` on Windows):

```
<!-- Example Linux Log Ingestion Configuration in ossec.conf -->
<localfile>
  <log_format>syslog</log_format>
  <location>/var/log/auth.log</location>
</localfile>

<!-- Example Windows Event Log Ingestion Configuration -->
<localfile>
  <log_format>eventchannel</log_format>
  <location>Security</location>
</localfile>
```

4. FILE INTEGRITY MONITORING (FIM / SYSCHECK)

The **Syscheck** module monitors critical system files and web directories for unauthorized creation, modification, or deletion. It records cryptographic hashes (MD5, SHA-1, SHA-256), file permissions, ownership, and timestamps to detect tampering.

```
<!-- Example FIM Real-time Monitoring Block in ossec.conf -->
<syscheck>
  <directories check_all="yes" realtime="yes">/var/www/html</directories>
  <ignore>/var/www/html/tmp</ignore>
</syscheck>
```

5. VULNERABILITY DETECTION & ASSESSMENT

Wazuh incorporates a native Vulnerability Detector module. Agents send installed software package inventories to the manager, which cross-references package names and version numbers against official National Vulnerability Database (NVD) CVE feeds. This automatically identifies unpatched system vulnerabilities (such as CVE-2021-44228 / Log4Shell) without requiring external vulnerability scanners.

6. XML DETECTION RULES & DECODERS

Incoming logs pass through a two-stage evaluation engine:

1. **Decoders:** Extract fields (IPs, usernames, status codes) from raw log strings using regex matching.
2. **Rules:** Evaluate parsed fields against XML rule trees. Rules are assigned severity levels (0 to 16) and mapped to MITRE ATT&CK technique IDs.

```
<!-- Custom Wazuh Rule Example: Detects SSH Brute Force (Rule ID 100050) -->
<group name="syslog,sshd,">
  <rule id="100050" level="10">
    <if_matched_sid>5716</if_matched_sid>
    <same_source_ip />
    <frequency>8</frequency>
    <timeframe>120</timeframe>
    <description>Possible SSH Brute Force Attack Triggered from $(srcip)</description>
    <mitre>
      <id>T1110</id>
    </mitre>
  </rule>
</group>
```

7. ACTIVE RESPONSE REMEDIATION

Wazuh **Active Response** executes automated countermeasures on agents when specific rules or severity thresholds are triggered. For example, upon detecting an SSH brute-force attack (Rule 100050), the manager instructs the agent to automatically block the attacker's IP using `firewall-cmd` or `iptables` for 30 minutes.

```
<!-- Active Response Configuration in ossec.conf -->
<command>
  <name>host-deny</name>
  <executable>host-deny</executable>
  <expect>srcip</expect>
</command>

<active-response>
  <command>host-deny</command>
  <location>local</location>
  <rules_id>100050</rules_id>
  <timeout>1800</timeout>
</active-response>
```

8. KEY LEARNINGS & CONCLUSION

Wazuh provides a robust, open-source SIEM/XDR alternative that seamlessly combines host integrity monitoring, vulnerability auditing, log correlation, and active response. Mastering Wazuh architecture equips SOC analysts with comprehensive endpoint security visibility and automated incident response capability.